(1.) Course Title: INFPHD617 Data structures and algorithms (complex exam preparatory)	**Credits: 6**
Course **Classification: Mandatory chosen**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 80% theory, 20% practice	
Type of Class: lecture and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -	
Assessment Method (Exam / Midterm Grade / **Other**[3]): colloquium **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: -	
Course Placement in Curriculum (Semester Number): 4th semester (autumn)	
Prerequisites (*if any*): -	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

- Abstraction levels of data structures: abstract data type (ADT), algebraic and functional specification of the operations; abstract data structure (ADS), the graph of the structure, operations; level of representation: representing data by arrays and by linked lists (pointers).
- Arrays, stack, queue, priority queue and the heap; operations, representations and applications in algorithms (e. g. polish form, graph searching, heap sort, improving the effectiveness of graph algorithms).
- Linked lists; simple list and using sentinel record (head element), double linking, cycling by pointers; operations, representations and applications of lists.
- Binary tree, representation by array and with pointers; tree traversal algorithms (preorder, inorder, postorder and by levels); recursive functions on trees; applications.
- Binary search tree, operations and their efficiency. The AVL-tree; relation between the height and the number of nodes in the tree; rotations (by insertion); the cost of checking and supporting the balanced property.
- The 2-3 tree, basic operations, efficiency; splitting by insertion and merging nodes by deletion; B-trees on the external storage devices.
- Hash tables, hashing. Resolving the key collision by chaining. Resolution of the key conflict by open addressing; linear and quadratic probing, double hashing. Hash functions (the division method).
- Graphs, representation by adjacency matrix and by adjacency lists, their efficiency and applications. Elementary graph algorithms: breadth-first search, shortest paths, minimum cost spanning tree; depth-first search, classification of edges, topological sort.

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

- J. Kleinberg, É. Tardos: Algorithm Design. Addison-Wesley, 2006.
- T.H. Cormen, C.E. Leiserson, R.L. Rivest, C Stein, Introduction to Algorithms. Fourth Edition. The MIT Press, 2022.

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.1
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.

[1] **Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

• Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation. • Engages as an equal discussion partner with professionals in the field. • Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.
Course responsible (name, position, academic degree): **Dr. László Ferenc Szabó, Habil. Associate Profesor, PhD**
Instructor(s) involved in teaching the course (if any) (name, position, academic degree): -